

Predicting Smartphone Addiction Through Deep Learning

A Capstone Project report submitted

in partial fulfillment of requirement for the award of degree

BACHELOR OF TECHNOLOGY

in

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

by

K.MANUSH	(2203A51081)
M.VEDASREE	(2203A51371)
A.NITHIN CHANDRA	(2203A51701)
B.SHIVA SAHITHI	(2203A51718)
G.RAVITEJA	(2203A51721)

Under the guidance of

Mr. S. Naresh Kumar

Assistant Professor, School of CS&AI.



SR University, Ananthasagar, Warangal, Telangana-506371

SR University

Ananthasagar, Warangal.



CERTIFICATE

This is to certify that this project entitled “**Predicting Smartphone Addiction Through Deep Learning**” is the bonafied work carried out by **K.Manush, M. Vedasree, A.Nithin Chandra, B.Shiva Sahithi, G. Raviteja** as a Capstone Project for the partial fulfillment to award the degree **BACHELOR OF TECHNOLOGY** in **School of Computer Science and Artificial Intelligence** during the academic year 2025-2026 under our guidance and Supervision.

Mr. S. Naresh Kumar

Assistant Professor,

SR University

Ananthasagar, Warangal.

Dr. M.Sheshikala

Professor & Head,

School of CS&AI,

SR University

Ananthasagar, Warangal.

Reviewer-1

Name:

Designation:

Signature:

Reviewer-2

Name:

Designation

Signature:

ACKNOWLEDGEMENT

We owe an enormous debt of gratitude to our Capstone project guide **Mr.S.Naresh Kumar, Asst.Prof . Professor** as well as Head of the School of CS&AI , **Dr. M.Sheshikala, Professor** for guiding us from the beginning through the end of the Capstone Project with their intellectual advices and insightful suggestions. We truly value their consistent feedback on our progress, which was always constructive and encouraging and ultimately drove us to the right direction.

We express our thanks to project co-ordinators **Mr. Sallauddin Md, Asst. Prof.** for their encouragement and support.

Finally, we express our thanks to all the teaching and non-teaching staff of the department for their suggestions and timely support.

CONTENTS

S.No	Chapter/Section Title	Page No
1	Abstract	1
2	About the Organization or Institute	2
3	Introduction of Project	3
4	Related Work	6
	2.1 Introduction	6
	2.2 Existing Systems and Limitations	6
	2.3 Eye - Activity Monitoring Experiment In Real Time	7
	2.4 Lightweight Deep Learning Model Py Investigation.	7
5	Problem Statement	10
6	Requirement Study,Risk Study and Feasibility Study	11
	4.1 Requirment Analysis	12
	4.2 System Requirements	14
	4.3 Risk Analysis	15
	4.4 Feasibility Study	17
7	Recommendations Solution/Method/Technique	23
	5.1 Inroduction	23
	5.2 Overall System Architecture	24
	5.3 Proposed Methodology	25
8	System Architecture , Flow charts and Data Flow Diagrams	28
	6.1 Introduction	28
	6.3 Flow chart of the System	30
	6.4 System Design	32
9	Result and Discussion	36
10	Conclusion	39
11	Feature Scope	40
12	References	42

LIST OF FIGURES

Fig No.	Title of Figure	Page No.
1	Flow Chart	30
2	Class Diagram	32
3	Home page	36
4	Dashboard Page Before Staring Application	36
5	Blink Time	37
6	Predicting Normal/ High	38

ABSTRACT

The project will focus on predicting and estimating the addiction to smartphones by applying the methods of deep learning by studying real-time eye activity instead of the traditional methods of statistics of phone use. Using the data in the form of algorithms to work with the characteristics of blink frequency, duration, variability, and eye-state classification, in conjunction with the data regarding user sessions will allow the system to go beyond simple screen-time metrics and capture behavioural and cognitive indicators of overuse. Using live camera feeds and model lightweight landmark finding and eye state classification models such as Retina Face and MobileNetV2, the project assembles a complete pipeline that identifies features and deferrals them to a multilayer perceptron (MLP) to predict the estimate degree of addiction risk.

The methodology will consist of Data preprocessing, eye-region cropping, normalisation, feature engineering and model training and evaluation. Various machine-learning methods, such as the classification of eye states, and regression-type predicting risk scores are compared to identify the most effective method to predict smartphone addiction. The system also provides prompt feedback and recommendations given in a personalised manner to promote healthier screen habits.

On the whole, the project makes a contribution to the expanding sphere of digital-wellness analytics by utilizing artificial intelligence and machine learning to forecast the risk of addiction to the smartphone and, thus, develop the body of knowledge concerning the relevance of using the mobile phone in a healthy way and allowing people to maintain healthier relationships with their technological devices.

ABOUT THE ORGANIZATION OR INSTITUTE

This project was developed under the **School of Computer Science and Artificial Intelligence (CS & AI) at SR University, Warangal**, a leading institution known for innovation, research, and experiential learning. The university fosters creativity, entrepreneurship, and technical excellence, enabling students to transform ideas into impactful solutions.

The **Capstone Project Program** encourages students to apply research and technology to solve real-world challenges. Under faculty mentorship, teams design and develop intelligent solutions that address industry and societal needs.

The project “**Predicting Smartphone Addiction Through Deep Learning**” integrates **Computer Vision, Behavioural Analytics, and Lightweight Deep Learning Models** to assess smartphone addiction using real-time eye-blink monitoring. The system captures live video, analyzes blink behaviour using RetinaFace and MobileNetV2, extracts meaningful statistical features, and predicts addiction levels through a Multilayer Perceptron (MLP) classifier. This project contributes to the growing field of digital wellness by providing a scientific and automated method to monitor overuse and fatigue associated with prolonged smartphone usage.

This work was completed under the mentorship of **Mr.S.Naresh Kumar**, Assistant Professor, School of **Computer Science and Artificial Intelligence, SR University**, whose guidance and expertise were crucial to the project’s success

Team Number: 195

Team Members:

K.MANUSH	(2203A51081)
M.VEDASREE	(2203A51371)
A.NITHIN CHANDRA	(2203A51701)
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CHAPTER 1 – INTRODUCTION

1.1 OVERVIEW

Smartphone addiction is a behavioural problem that is currently fast gaining traction amongst the digitally-interconnected society in the current generation, specifically with the segment of the student population and young adults. The excessive use of the smart phone can lead to failure to be productive, sleeping difficulties and emotional addiction. Smartphone addiction which is self-reported measures or even crude ones such as screen time which in most instances cannot control real time behavioural patterns and signs of cognitive overuse.

The proposed solution to these inadequacies is the project called Smartphone Addiction through Deep Learning, which implies an intelligent and automated app along with the discovery of the potential of the smartphone addiction through Deep Learning and eye trackers live. This system does not solely depend on the statistical data on the use, but also examines the less apparent physiological features of the rate of blinking, the length of the blink, irregularity of the patterns of blinking and eye state. These parameters are computed on the lightweight face detectors of the landmark face such as RetinaFace and MobileNetV2 that can track eye-region when low-end computing is used.

The obtained features are inserted into machine learning pipeline, which includes of preprocessing, eye-region cropping, normalization, feature engineering and classification. Multilayer Perceptron (MLP) is trained to give the estimated risk of addiction, in fact by the real time behavioural response. The user session data is also crunched and compared to the eye based data and meaningful predictions are drawn by the system that shows the cognitive cognition and disproportionate usage trends of a user.

The interface can also offer the real time feedback to the users on the amount of danger and provide the less harmful patterns of usage. It is a strategy founded on the concept of deep learning and goes beyond more conventional measures of screen-time and offers a more realistic and behaviourally sound concept of smartphone addiction.

Overall, the project can be applied to the contemporary digital wellness analytics due to its ability to predict smartphone addiction with the great accuracy of the accuracy high level and in a personalized manner, with the assistance of highly developed AI models. It works together with helping individuals to adopt healthy screen patterns and creation of a healthy relationship with technology.

Objectives of the Project

The most important objective of the project is to create AI-based application, which will be able to predict the addiction to the smartphone, basing on the real time eye pattern and deep learning. The major objectives are:

Development of a video processing pipeline, which records the eye movement of the camera feed in real time.

Eye-state recognition and localization on RetinaFace and mobile net V2 models.

The frequency, duration and variability of the behavioural characteristics of the blink were used to carry out the pull out with an aim of carrying out the addiction analysis.

Frame cropping of the eyes, normalization and feature engineering.

Multilayer Perceptron (MLP) model Coding and development to predict addiction risk.

Comparison of different methods in machine learning as well as the desire to know which of them would be the most appropriate in terms of prediction accuracy.

The availability of either a personalized feedback and recommendation to the users based on the perceived risk of becoming addicted.

Rapid, accurate and on time analysis, which can be applied in day-to-day basis.

1.2 SCOPE OF THE PROJECT

The project plan will consist of real-time behavioural analysis, addiction prediction, enhancing the digital wellness, and offering feedback to the user. It targets the audience, scholars in the mental health community, and digital wellness by providing an AI-based solution to learn smartphone addiction.

For individuals, the system:

- It has real time eye- blink pattern monitoring and cognitive load monitoring.
- Deems the risk issue of addiction to behavioural measures and not on the self-report questionnaires.
- Recommends himself, personally, the ways of decreasing the ill use of smartphones.

The system:

- Gathers behavioural information founded on monitor eye-states.
- Presents the opportunity to analyze digital wellness via the objective and AI-driven measures.
- Helps to analyze cognition and physiological predictors of heavy smartphone use.
- The structure of the project is of modular nature and due to this, it is possible to:

- Association with the mobile applications or association with digital wellness.
- Expansion into other capabilities such as emotion or attention tracking.
- Population giant analytics make their massive entry into wellness.

The behavioural records are anticipated to be recorded and plotted in an efficient form of maps and model output that is arranged. The combination of deep learning, behavioural analytics and real-time surveillance would provide a platform that is robust and capable of scaling, and the project could be applied to address the dynamics of the digital wellness and addiction studies.

1.3 SIGNIFICANCE OF THE PROJECT

The modern environment of technologies is likely to make individuals unable to recognize the first signs of engaging in smartphone misuse. The previous forms of measurements are unable to provide real time behavioural data and therefore, it results in latent awareness and inability to properly manage overuse.

The Deep Learning-based system offered can be of significant relevance to the process of enhancing the evaluation process as it provides:

- Blink-pattern and eye-state analysis Automated real-time.
- Prediction of addiction behaviour is done through an existing neural network.
- The personal feedback which encourages a better usage behavior.
- This will enable individuals to understand they are addicted to their smartphone using objective measures and not assumptions and self-reports.

The system will be an efficient analytical tool to the researcher and wellness practitioners because it will present:

- The exact behavioural symptoms of digital addiction.
- Concepts regarding user engagement behavior, cognitive burnout and overuse.
- Promotion of the development of online wellness programs that are tailored.

The new lines of research that are brought about by the project in the area of artificial intelligence-based mental health, behavioural surveillance, and personalised control of online habits are also new. The architecture is ethical and assures privacy, openness, and security to the users; it also informs of the trends of smartphone use.

CHAPTER 2 – RELATED WORK

2.1 INTRODUCTION

Live eye-activity recording has proven to be a powerful tool of insights of the human behaviours, cognitive burdens, exhaustion and attention patterns. Drowsiness, stress and excessive use of computers are some of the mental states that can be identified with the introduction of the computer vision and deep learning. Conventional methods of identifying smartphone addiction are based on the concept of self-reported questionnaires or logs based on screen time, which cannot identify patterns of actual behaviour.

Smartphone Addiction With Deep Learning, is an architecture that combines Metric RetinaFace, MobileNet V2 and blink-timeline modelling and an MLP-based prediction engine in finding out the degree of addiction of a person based on the physiological and behavioural characteristics.

The chapter is a literature review of the literature on the eye-activity monitoring, lightweight deep learning trainers, blink-pattern detector, feature extraction techniques, and behavioural state predictor models. The weaknesses of the present systems are discussed and how the proposed system will address the issues based on a light weight, light weight and behaviour based architecture are elaborated in a real time.

2.2 EXISTING SYSTEMS AND LIMITATIONS.

The current systems condition in the eye-tracking and behaviour monitoring should be associated, more or less, with the fatigue detection, the driver drowsiness, or with the medical surveillance. Their success in their respective fields has not been mentioned but predicting addiction to Smartphones.

Weaknesses associated with the current systems:

None of the real-time addiction solutions is provided - The current solutions are able to identify drowsiness or fatigue but not digital addiction or overuse behaviour.

Too large to fit on the phone- CNN-LSTM hybrid or even the large models that are not realistic to run on everyday smartphone application.

None of the blink-timeline modelling Systems describe the rate of blink but none of them often considers such variations as variability, frequency peaks or prolonged concentration.

Not all the features are projected to the addiction indicators Blink duration, blink irregularity, and eye-state imbalance do not belong to the addiction environment.

Lack of feedback systems- The existing systems lack individualized wellness notifications or forewarned dangers.

These shortcomings suggest the necessity to possess a light, real-time, and behaviour-driven framework that can be applied to smartphone-supportive models like RetinaFace and MobileNetV2 alongside MLP-based predictor.

2.3 EYE - ACTIVITY MONITORING EXPERIMENT IN REAL TIME.

The effectiveness of the surveillance based on the eyes in the analytical process of the behaviour is proved by a variety of researches:

Sharmila et al. (2019)

ViolaJones Blink frequency to identify Computer Vision Syndrome. Their experiment confirms this theory because the rate of blinking by individuals is a measure of mental stress.

Rajamohana et al. (2021)

Created CNN + BiLSTM algorithm to track the eye condition and blink of drivers. This is in favour of real time eye-state classification that is applied in MobileNetV2.

Quddu et al. (2021)

The applicability of time-related blink sequence analogous to the Blink Timeline element has got to do with blink sequences as the foundation of classifying drowsiness, grounded on RNN and LSTM.

Liu et al. (2020)

Perceived fatigue using applied PERCLOS (percentage of eye closure) that made the conclusion that the time of the eye-closure is an effective predictor of behaviour - generalises to addiction modelling.

Wang et al. (2019)

Proposed a two-stream CNN with gaze and pupil tracking, as by doing this, one realizes the significance of proper landmark localization, just like in the case of RetinaFace.

Such studies are the scientific principles of real time observation of the deduction of blinking and behaviour.

2.4 LIGHTWEIGHT DEEP LEARNING MODEL PY INVESTIGATION.

Most researchers examine efficient architectures in order to make the performance mobile-friendly:

The landmarks that can be extracted by retinaface are extremely specific and quick, and at low prices of computer computation.

MobileNetV2 is common in light-weight classification models and it has been demonstrated to be effective in the eye-state classification.

In comparison with heavy networks (e.g. ResNet or VGG), the lightweight CNNs are better according to real-time work on mobile devices as per research.

This is why the proposal of RetinaFace + MobileNetV2 is suggested.

2.5 BLINK-TIMELINE AND BEHAVIOURAL PATTERNING

RESEARCH.

The history indicates that the conditions of behaviour are associated with the modes of the blinking:

Large discrepancy in blinking is among the symptoms of mental exhaustion or reduced concentration.

It is slow rate of blinking at very long intervals of time that is a sign of absolute focus or obsession.

The non-patterned blink code is associated with the lot of time spent on the screen.

These notions coincide with the Blink Timeline (per 2 minutes) course of your system.

2.6 FEATURE ADDICTION INDICATORS AND EXTRACTION

Based on prior literature:

Blinking frequency ▶ 3: degree of engagement.

Blink time - is a pointer to fatigue or mental strain.

Compulsive digital behaviour is linked with Blink variability -

Eye-state ratio- assists in formulating the levels of attention.

It is these that are excised in your system to form a set of features correlated to the behavioural markers of addiction.

2.7 MLP-BASED PREDICTION MODELS

They are used in the MLP (Multi-layer Perceptrone) models.

behavioural classification

risk scoring

cognitively regress the indicators.

As has been shown, MLPs can be beneficial in the case, provided the features represented by the vectors are clumped and predictive and, thus, can be of relevance in estimating the addiction-risk.

Feedback: Direct feedback will assist the workers to rectify their quality of work since it is an also a conversational process that is characterized by sharing insights, information and aiding feedback with the workers.

Recent publications on digital wellness are in favour of:

- instant feedback
- personalized alerts
- behaviour-based recommendations

However, the systems being used nowadays cannot accommodate physiological real time monitoring. This empty space is filled by your proposal to design include:

- a real-time alert system
- personal risks (Low / Medium / High)
- suggestions as to behaviour of action.

This makes this system holistic when compared to any fatigue based or a screen time based models.

Carrying out research is required to identify the research gap.

The huge gaps founded on the literature review are:

To anticipate the smartphone addiction, no technology will take into account the real-time eye tracking.

The deep learning models are cumbersome, which they cannot be used in real-time.

Digital addict analysis Blink-timeline behaviour Blink-timeline behaviour has received a limited amount of research on the subject of digital addiction.

All systems of feature extraction do not produce eye-based signals as corresponding to addition predictors.

The existing fatigue / drowsiness solutions do not have a personalized wellness feedback.

CHAPTER-3 – PROBLEM STATEMENT

The long-term use of smartphones has gradually turned into a cause of digital addiction, stress, and lack of concentration with a harmful effect on cognitive abilities and overall productivity. The more time people spend on their gadgets, the more they tend to overlook the fact that their habits of use affect the mental state. The self-regulation process is hard and many users have no clue about the subtle signs of overuse and their behavior, and thus unhealthy habits develop without any significant notice.

Current smartphone wellness apps do not measure anything beyond the screen time or app usage. Although they are helpful, these measures do not give a vivid picture of the online activity of a user and cannot define the degree of addiction accurately. They do not have the capacity to study more behavioral clues related to compulsive smartphone use.

A deep learning-based system can be significant to fill this gap. With the ability to capture intricate behavioral patterns and suggest meaningful insights by examining them using advanced models, such a system will be able to classify addiction levels more accurately. Moreover, it is capable of delivering tailored suggestions that will lead users to healthier online lifestyle choices, fostering the improvement of their mental health and more moderate smartphone usage.

CHAPTER 4 : REQUIREMENT STUDY, RISK STUDY AND FEASIBILITY STUDY

Project requirement and feasibility investigation should also be developed before any project requirement. The Smartphone Addiction through Deep learning project is driven by the eye-activity monitoring (real time) in an effort to identify the level of addiction of an individual in allusion to the eye-activity patterns (blinking, eye-state recognition and behavioural patterns). The chapter is the description of technical requirements, limitations of the work with the system, risks, and the probability of the system development. It studies the interface of the system and its connection with the hardware, software and the user to confirm that all the modules of the system such as the video frame acquisition to the alert system is both technically and viability viable.

The chapter is divided into big sections as follows:

- Functional Requirements - The definition of such major features as the eye recognition, eyelid tracking, features extraction, and addiction risks prediction.
- Non- Functional Requirements- The displaying of performance, reliability, usability, responsiveness and security requirements.
- Risk Analysis-Determining the technical and operation issues and how they can be solved.
- Feasibility study -Technical, operational and economical feasibility shall be studied to be put into practice.

The result of the synthesis of these analyses will give the system a big picture and will attract a prototype of the design of the stable and real-time predictive system of addiction.

4.1 REQUIREMENT ANALYSIS

One of the tasks, on which the proposed system will be proven to be helpful, is the requirements analysis, which condenses how the system is expected to work, how the various components will interact and provide the necessary real-time forecast of addiction. There are Non-Functional and Functional requirements.

4.1.1 FUNCTIONAL REQUIREMENTS

Functional requirement Statement shows the most applicable processes that ought to be delivered by Smartphone Addiction Detection System. These functions are directly associated with stages of flowcharts in the system.

1. Video Frame Input

The system should be capable of recording video frames in live video of a web-cam or a video recording.

- Frames have to be dynamic (60-30 FPS would be a good idea).

2. Face and Eyes recognition (RetinaFace Module)

- RetinaFace must detect:
- Full face region
- Eye landmarks
- Restricting three boxes of right and left eyes.

In steadicam eye cropping correct identification of landmarks in the proper world should be precise.

3. Eye Crop Extraction

The system will be required to detect the parts of the eyes and get them removed automatically. The MobileNetV2 requires eye crops that have been scaled and normalised.

4. MobileNet V2 (Eye State Classification).

The clusters of the eye crops will be:

- Open Eye
- Closed Eye

The classifier is supposed to make frame sensitive forecasts.

5. Blink Timeline Generation

The frequency of the blink per a given time (each time after every 2 minutes) will be identified in the system.

- It must store:
- Blink frequency
- Blink duration
- Blink irregularity

6. Feature Extraction

The system has to be ready to acquire the nature of behavioural peculiarities regarding the addiction:

- Blink frequency
- Eye-closure duration
- Blink variability
- Attention consistency

7. Addictions (MLP Model) Prediction.

The features extracted are required in an MLP classifier.

The system should be at a point of categorizing the addictions degrees as:

- Low
- Medium
- High

8. Alert System

In anticipation of when it has been done, the system needs to be ready to create:

- Visual alerts (color-coded)
- Brief explanations

Behaviour recommendations.

The non-functional requirements are the requirements that would be entailed in the manner the system would behave.

The non-functional requirements help in making sure that the system is of good quality responsive and reliable even during the real time.

1. Scalability

System is also going to be working with numerous predictions at the same time.
Supported on big data.

2. Security

There should be no patterns of blinking and predictions in any form of video form.
The logs should be encrypted/authenticated in such a way that it is barred by firewall.

3. Usability

Inconveniently: user interface must also be capable of:

- Starting analysis
- Viewing addiction level
- Interpretation of the findings using graphs.

4. Reliability

- There should be minimization of system crash.

Fallback mechanism is supposed to be activated when the frames are lost or the diagnosis is mistaken.

5. Performance

- Real-time detection under:
- 30–40ms per frame

- Low latency prediction

There is minimal inference and loading model to be done.

6. Portability

- Should run on:
- Windows
- Linux
- Mac
- Jupyter Notebook or VS Code

7. Maintainability

To exist: architecture must be modular because:

- Model updates
- Parameter tuning
- Adding new features

8. Data Privacy

- Must not store facial images.

The processed features of the blink are only saved on-demand.

4.2 SYSTEM REQUIREMENTS

System requirements are what and what software it will need to carry out all the behavioural prediction pipeline.

1. Software Requirements

JavaScript Figure 1 / Jupyter notebook / Google Colab.

To compose the code and model experimentation.

- Python
- GitHub for version control.

2. Packages and Frameworks

One will require TensorFlow / Keras at least to load and train MobileNetV2 and MLP.

Eye and face landmark Python Face and eye landmark detector (RetinaFace).

Computing NumPy Pipeline array.

- Piped computations of numbers arrays.

Pandas - It involves establishment of creation and attribute agenda.

Graphs of the rates of blinking and addiction in Matplotlib / Seaborn.

classification-1 identification of addiction scikit -Learn - Extraction, evaluation, final classification.

3. Hardware Requirements

- Laptop/PC with:
- Webcam
- Minimum 4GB RAM
- Quickening CPU/ GPU inference.

4.3 RISK ANALYSIS

Risk analysis is one of the tools that may be adopted; it measures all the challenges that might be experienced in the development and deployment.

4.3.1 DATA-RELATED RISKS

1. Incomplete or Biased Data

- Blink has not lots of demographics it allows.
- Mitigation: Research patterns of various age and screen time group.

2. User Behaviour Variability

- The variables which are considered to define the pattern of the blinking are the light, stress and fatigue.
- Mitigation: Problem (uniform light, face visibilities) instruction applications.

3. False Feature Extraction

- The distortion of the nature of the blink may occur after the displacement of the lost frames or absence of the eye position.
- Mitigation: No noise Frame validation or reprocessing should be added.

4. Landmark Detection Errors

- Retinaface cannot be used in the dark.
- Remediation: Recommend such a user to turn on lights.

5. Data Loss / Corruption

- Everything that Blink has accomplished as regards to forecasts or projections is not archived.
- Payments: Gilt lumbering and improper storage.

4.3.2 TECHNICAL RISKS

1. Overfitting / Underfitting

- Mitigation: Use:
- Cross-validation

- Dropout
- Learning-rate optimization

2. Feature Irrelevance

- Mitigation: Choose the properties to blinking.

3. Pipeline Breakdown

- Alternative Fall back modules (per stage).

4. Hardware Compatibility

- Mitigation frame rate and webcam resolution should not be less than 20.

4.3.3 PERFORMANCE RISKS

1. Video Processing Latency

- Low-end devices may lag.
- Mitigation MobileNet V2(small) and downsize frame.

2. Model Loading Delay

- Cache model weights.

3. System Downtime

- Mitigation: Sign crash recovery and monitoring services.

4.3.4 OPERATIONAL RISKS

1. Deployment Issues

- Mitigation: It involves complete environment establishment.

2. Maintenance Challenges

- Mitigation Refreeze models differences.

3. User Engagement Risks

- UI is sluggish or is bulky to the users.
- Mitigation: better instructions and colour coded maps.

4.3.5 SECURITY RISKS

1. Privacy Exposure

- Facial frames may leak.
- Mitigation: do not store raw video, blink data.

2. Unauthorized Access

- Mitigation: AX: safe logging identification and prediction.

3. Malware / Code Tampering

- Rebate: personal GitHub plan.

4.3.6 LEGAL & ETHICAL RISKS

1. Various regulatory violations in violation of privacy.
 - Minimization interventions: Privateness and consent policy.
2. The false depiction of the Addiction levels.
 - Mitigation: Disclaimers and statements of issues.
3. Ethical Use of Facial Data
 - Mitigation Do not allow the use of unreasonable data in the system

4.4 FEASIBILITY STUDY

1. Technical Feasibility
 - As a model that will be used in real-time, lightweight models (RetinaFace + MobileNetV2 + MLP) will be used.
 - These laptops are also wireless and of mid range.
2. Operational Feasibility
 - Little contact with the user, little UI.
 - Naturally, among the laymen, among the students or wellness academicians.
3. Economic Feasibility
 - Open/Free software (Python, Tensorflow, OpenCV).
 - No costly hardware required.

4.5 FEASIBILITY STUDY

The viability analysis can confirm that the Smartphone Addiction Detection System based on the Deep Learning technology can be developed, designed, and implemented in the perspective of the technical, financial, operational, and ethical factors. It will look at the likelihood of the real-time eye-activity monitoring system and the sustainability in terms of the availability of resources, the complexity of implementation process and viability of the system, or not, in the long-term.

The result of this study would be a system that is technical and economically viable that would benefit the individuals who will be using it like students, employees, researchers and wellness practitioners. The analysis at the diversified levels of the feasibility would assist the project team in arriving at the conclusion that the system per se would be deployed successfully, serviced and extended with the minimum levels of risk and a maximum spectrum of performance.

The proposed system feasibility test is done on the premise that the following are the key sectors:

1. Technical Feasibility
2. Economic Feasibility
3. Operational Feasibility
4. Ethical & Legal Feasibility
5. Market Sustainability & Feasibility.

All the dimensions are addressed in order to make sure that viability of the whole system is considered.

4.5.1 TECHNICAL FEASIBILITY

The Smartphone Addiction Detection System could be a viable creation model, since it is deployed using lightweight, effective and ubiquitous deep-learning, which can be run on the standard computing equipment without any inconvenience. Eye-landmarks and eye-states are recognized by the two significant constituents of the model, RetinaFace that identifies faces and eye-landmarks and MobileNetV2 that identifies eye-states and does not require considerable amount of computation power.

The face will guarantee that the face-surface and eye-surface will be recognized under the subsequent conditions:

- Partial face rotation
- Low lighting

Semi-occlusiveness or eyeglasses.

On the one hand, the MobileNetV2 is inverted residual and the implementation of the linear bottlenecks that classify the conditions of human eyes (open/ closed) and, thus, can be used by any type of device laptops, tablets, and smartphones.

The system will be constructed using python and open-source deep-learning frameworks like TensorFlow/Keras, NumPy, Pandas and openCV so that the system could be flexible, reproducible and scalable.

It is in a modular form and consists of:

- Video frame acquisition
- RetinaFace detection
- Eye cropping
- MobileNetV2 classification
- Blink timeline generation

- Feature extraction
- MLP prediction of addiction.
- Alert system

The elements are developed case-by-case, that is, they can be brought to even newer versions (e.g. replacing MLP with LSTM/Transformer) without affecting other elements of the pipeline. The system does not need any special eye tracking equipment like use of infrared cameras or expensive sensors. It just requires an inbuilt web camera that would render any technical implementation simple and scalable.

On the whole, one can say that the system is not something beyond the reach of the existing technologies, the resources available are enough to make it work and it is not a challenge to implement it in the educational context, organizational context, and the wellness sphere.

4.5.2 ECONOMIC FEASIBILITY

The proposed system is economically feasible due to the use of free and open-source technologies, avoiding expensive licensing fees or proprietary hardware.

Cost Components Overview

Component	Estimated Cost (INR)
Development, Testing & Maintenance	₹1,00,000 – ₹1,50,000
Computing Resources (local/cloud)	Minimal (Colab GPU free tier available)
AI Libraries & Frameworks	₹0 (Open-source: TensorFlow, NumPy)
User Interface/Visualization	₹20,000 – ₹30,000

Key points:

- The system can be used with normal webcams - which can be found in laptops or desktops - they do not require any additional hardware installation.
- Training and testing: the training and testing can be done with the assistance of Google Colab where GPUs and TPUs are free.
- An opportunity to apply no special sensors or paid APIs.
- Institutions and individuals who have low investments can adopt the system.

This renders the system highly economical, scaled and fit to be utilized in establishments of learning, research laboratory, health facilities and working environments.

4.5.3 OPERATIONAL FEASIBILITY

It is operationally highly feasible system because of automated workflow and good user friendliness. The users are not demanded to have any technical expertise in the usage of the real-time detection pipeline.

Ease of Use

Users only need to:

1. Sit in front of a webcam
2. Ensure proper lighting
3. Within some minutes, the system is able to keep track of the eye activities.

Background is also automatically involved in identifying the features of the blink detection, prediction and alert generation of addiction.

Modular System Operation

Each of the modules (Retinaface, MobileNetV2, MLP classifier and blink feature extractor) is independent.

In case any of the components is changed, the rest of the components will not be interfered.

1. Maintenance is made easier and standardized.

Adaptability

The system can be deployed in:

- Colleges and universities
- Workplaces
- Health and wellness centers
- Digital well-being programs

Research and behavior laboratories.

The reliability exhibited by the system, simplicity of implementation and the low interference of the users exhibited by the system make it operationally viable in other settings.

4.5.4 ETHICAL & LEGAL FEASIBILITY

The system is consistent with the ethical and legal principles of the facial data, privacy of the users, and responsible behavior of the AI.

Privacy and Data Protection

- Like no uncooked face-images, or video frames.
- Prediction of addiction is performed temporarily using the characteristics of numerical blink.
- No personal identifiable information (PII) will be stored.
- Video frame capture shall only be permitted on the consent of the user.

The risk of the cloud privacy can be eliminated by the means of the carrying out of all the processing on-site.

Legal Compliance

- Complies with privacy policies including GDPR-like.
- Keeps to a minimum the data and purpose processing.
- It is stored in non-biometric format, in other words it is not so complex as per the law.

Ethical AI Practices

- The system is not biased as it considers only the physiological peculiarities of blinking.
- An explanatory and clear description is provided of the state of addiction.

Production is not evaluated or discriminated but it is well.

The system is law-abiding and ethical that should be applied in the institutions of learning, organizations, and wellness.

4.5.5 MARKET FEASIBILITY AND SUSTAINABILITY

The increase in the number of smartphone users population, pupils, employees, adolescents makes the demand of the scientific and plausible digital wellness solutions grow. The existing technologies in the field of well-being are also constrained as they can only monitor the screen time but not physiological signs of addiction.

Market Gap

There is a lack of:

- Addiction identifying biometric.
- Live wellness biometrics.
- Eye-activity pattern scientific assessment tools.
- Advantages of the Proposed System in the market.

- With bio-data, rather than self-report.
- Can be mounted on the normal machines.
- Funds digital mental and health initiatives.
- Very scalable to very large number of users.

Sustainability

- The system is still applicable because of:
 - Amplifying online reliance.
 - Personal surveillance requirement, mental-health.

Opportunity to be upgraded (stress analysis, emotion detection, etc.) in the future.

It is economical, highly scalable and scientific therefore making it long term viable in the market of digital wellness.

CHAPTER 5 :

RECOMMENDATIONS SOLUTION/METHOD/TECHNIQUE

5.1 INTRODUCTION

Subjective questionnaires, screen-time logs and self-reports are the conventional methods of quantification of the addiction systems of smartphones. These processes are not predispositions to immediate behavioural patterns and physiological indications that imply forcible smart phone use. The when time is also a sign, however, this can also be a sign of addiction and the cognitive load and the involuntary eye-blink as such, thus, the traditional method of assessment proves inaccurate, not to mention that they did not offer individual feedback to the users.

That is why, the Deep Learning Smartphone Addiction Prediction System is aimed at eliminating such shortcomings with the help of the computer vision and eye-activity real-time tracking. The suggested solution will be the shallow learning with the deep neural networks to detect the eye states and analyse the blink behaviour and determine the degree of addiction by employing the behavioural biomarkers. The system records dynamic signs contrary to the documentation of the stationary, thus the frequency of the blink, irregularity and eye-close patterns among others are dynamically recorded scientifically and which have been scientifically attributed to cognitive fatigue and excessive screen time.

The solution would imply the presence of a single real-time pipeline that would be composed of the computer vision, landmark detection, the eye-state classification, and the behavioural prediction that would be founded on the machine learning. This model has been trained on Python, RetinaFace, MobileNet V2, OpenCV, TensorFlow/Keras and webcam (ordinary) input. The platform carries out frame-wise recognition, feature recognition, addiction classification and live warning.

The deep learning and behavioural analytics will make the assessment of smartphone addiction a scientific, automated and end-user based approach that will assist in implementing a more healthy digital behaviour and will allow implementing the intervention in the case of the digital overuse.

5.2 OVERALL SYSTEM ARCHITECTURE

The overall system layout is the following one:

The architecture of Smartphone Addiction prediction System would be streamlined to attain the degree of real time, scalable and modular. The different modules are synchronized with each other and each component of the pipeline has a very narrow task which would provide some degree of predictability and usability.

The key layers of the proposed architecture will be the following:

1. Video Data (Input Module): Acquisition Data Layer.

- It is one of the layers that captures video frame of any phone camera or web-camera.
- It is also possible to process images in which resize, denoising and normalization are used.
- The original quality of the image is recovered as the subsequent computer vision operations are performed.

2. Face and Eye recognition, retinaface Engine.

- Overall, RetinaFace makes a point in which the face is found and the right patterns of the eyes are broken down.
- Adjusts to the change of light, direction and angles of face.
- The components of the eyes are extracted and sent to the classifier.

3. MobileNetV2 Layer (Classification of the eye-state).

- The MobileNetV2 is able to detect and indicate whether all eye crops are open or closed on frame-by- frame basis.
- It is light-wise i.e. it has the capability of giving real time classification and low latency.

4. Blink Construction Timeline Layer.

- The undetermined eye disorders are separated into time-series sliding windows (around 2 minutes).
- Timeline records:
- Blink count
- Blink duration
- Blink variability
- Eye-closure patterns
- This creates time basis behavioural pattern and not cases.

5. Feature Extraction Layer

- Blink timeline has behavioural features generated:
- Mean blink rate
- Maximum blink duration
- Irregularity index
- Total session duration
- All these are physiological signs of the digital fatigue and even overuse.

6. Addiction Layer (MLP Classifier)

- The rate of addiction is predicted using an MLP as follows:
- Low
- Medium
- High
- The model dictates the trends that should be pursued in case of excess of screens or saturation of mind.

7. Alert & Feedback Layer

- Whereas the prediction shows that it is Medium or High addiction:
- An alert message is then created in due time.
- The users would be advised to close their eyes or to spend less time in the screen or to take a break.
- The scalability of other platforms and flexibility and ease with which individual components can be modified are provided by the modularity.

5.2 PROPOSED METHODOLOGY

The methodology outlines the complete workflow used in the proposed system, from video acquisition to addiction prediction and user alerts. The system follows a structured, step-wise pipeline aligned with the real-time processing requirements of behavioural monitoring.

Step 1: Video Frame Acquisition

- The webcam or mobile camera streams live video input.
- Frames are captured at regular intervals for continuous monitoring.

Step 2: Preprocessing of Frames

- Preprocessing includes:
- Resizing frames
- Noise reduction
- Normalization

Step 3: Face Detection and Eye Landmark Localization

- RetinaFace accurately detects facial boundaries and locates eye landmarks.
- Eye regions are cropped using bounding boxes derived from the detected landmarks.
- Eliminates irrelevant background information to focus solely on eye behaviour.

Step 4: Eye-State Classification (Open/Closed)

- Using MobileNetV2, each cropped eye frame is classified as:
- Open, or
- Closed
- MobileNetV2 is chosen due to:
- Extremely low computational cost
- High accuracy
- Real-time inference capability

Step 5: Blink Timeline Generation

- The predicted eye states are arranged into a two-minute sliding window to build a blink timeline.
- This timeline captures:
- Number of blinks
- Duration of each blink
- Variability between blinks
- Long closures indicating fatigue
- The blink timeline forms the behavioural history necessary for addiction analysis.

Step 6: Feature Extraction

- Statistical features extracted include:
- Average blink frequency
- Maximum blink duration
- Blink irregularity score
- Eye-closure percentage per session
- These metrics serve as quantifiable indicators of digital fatigue and overuse.

Step 7: Addiction Prediction Using MLP

- The extracted features are fed into an MLP classifier, which determines the addiction level:

- Low Addiction – Normal blink patterns
- Medium Addiction – Noticeable irregularity or strain
- High Addiction – Strong signals of cognitive fatigue and overuse
- The classification is based on behavioural patterns rather than screen-time logs.

Step 8: Alert Generation and Feedback

- If the MLP model predicts Medium or High addiction:
- A notification is displayed alerting the user.
- Recommendations include resting, reducing screen exposure, or practicing digital wellness habits.

CHAPTER 6 – SYSTEM ARCHITECTURE, FLOW CHARTS, AND DATA FLOW DIAGRAMS

6.1 INTRODUCTION

The needs of systems architecture and data flow diagrams are important to know how the various modules of a software system interact, process information and how they communicate between themselves. These charts provide a pictorial representation in the internal workflow of the system, the stage of computation and data flow of such a system giving a clear understanding of how the system operates between the input and the output.

In the case of the Smartphone Addiction Detection System using Deep Learning, the diagrams are used to explain how the main components of the system, which include the video capture unit, preprocessing unit, RetinaFace-based face detection, MobileNetV2-based eye-state classifier, blink timeline unit, feature extraction unit, MLP prediction model, and alert system interact. All these elements collectively constitute an end to end behavioral analysis pipeline which is intended to be run in real time.

This chapter shows the system architecture, the interaction structure of the modules, and the data flow model of the proposed system. The figures illustrate the way of capturing and processing the live video frames taken by the user to identify the presence of eyes, the possibility of a blink behavior, their statistical features, and the ultimate evaluation of these features to identify the degree of addiction risk with the aid of an MLP classifier.

These graphic illustrations give a holistic picture of the work process of the system - starting with raw video feed and ending with real-time addiction notifications. They also bring out the modularity, extensiveness and scalability of the system, which demonstrates how the individual modules interface with each other to provide real-time, efficient, and accurate behavioral predictions.

6.2 SYSTEM PIPELINE AND FUNCTIONAL FLOW

The architecture of this system was done after considering two basic principles. The Smartphone Addiction Detection System via Deep Learning is developed in the form of the system architecture of a multi-stage pipeline, which works in real-time and uses video capture, computer vision, deep learning models, behavioural features extraction, and addiction levels prediction. The architecture begins with the Video Acquisition and Preprocessing Layer in which the live video frames are being captured by a webcam or smartphone camera all the time. These frames are preprocessed with resizing, noise removal operation and normalization

of frames to ensure a uniform quality and stability across the rest of the subsequent processing. This standardization is required in the development of a better face and eye detector module.

Following the frame preprocessing, the frames are subsequently sent to the Face Detection and Eye Landmark Extraction Module that is propagated by the RetinaFace deep learning model. In extreme conditions such as in the dark, off-angle shots or where the faces are partially covered, RetinaFace is applicable in localizing and detecting relevant features of the eye such as in the dark. In such landmarks, the system then proceeds to delete the regions of the eyes in each frame and once more provides further normalization to extract only the relevant visual data that is relevant to the blink behavior.

Eye-State Classification Layer examines the interior parts of the eyes and applies the MobileNetV2 model, which is lightweight. In this model, all the eye frames are classified as either an open frame or a closed frame (per frame). MobileNetV2 can be used on a computer of the commodity level under real-time inference, unlike high-end GPUs due to its optimized architecture. It can be able to recognize eye-state in fast and accurate manner and can process continuous videos smoothly.

The eye classified states in turn are grouped in the Blink Time-Construction-Layer, and arranged into a sliding time series window of two minutes. The significant behavioural patterns whereby the blink timeline captures are the rate of blinking, length of a blink, abnormality and eye-contact pauses. This kind of trends suggests the cognitive load and the degree of fatigue and the potential indicators of overscreening with which the user has been subjected.

Once a timeline is generated, the system passes on to the Feature Extraction Layer where the meaningful statistical and behavioural measures are determined. These are average rate of blink of the user, maximum blink, variability of blink time and average percentage of the eye-closure throughout the session. The features extracted generate a complete behavioural profile, and which represents the physiological response of the user towards screen use.

Its features are then sent to the Addiction Prediction Layer that uses Multilayer Perceptron (MLP) classifier. The MLP compares the trends in the blink behaviour and describes the degree of addiction as Low, Medium and High. With such a classification, the system will be able to detect normal behaviour, moderate strain, or signs of significant digital fatigue or even an addiction of the user to the smartphone.

Finally, there is the system concluding with the Alert and Feedback Layer where real time messages are displayed based on the estimated addiction level. The system delivers immediate messages to verify his/her screen time, take a break, or close his/her eyes in case

the user is registered on a Medium or High level of addiction. It is a feedback mechanism that can contribute to the improved usage of the device and support digital health.

6.3 FLOW CHART OF THE SYSTEM:

The flow chart has displayed the process of the system to process live video in order to identify smartphone addiction. It captures the camera feed, processes eye blinks with a deep learning neural network, gets behavioural features and ultimately predicts the addiction level. Information is given to the other step to produce a real time alert.

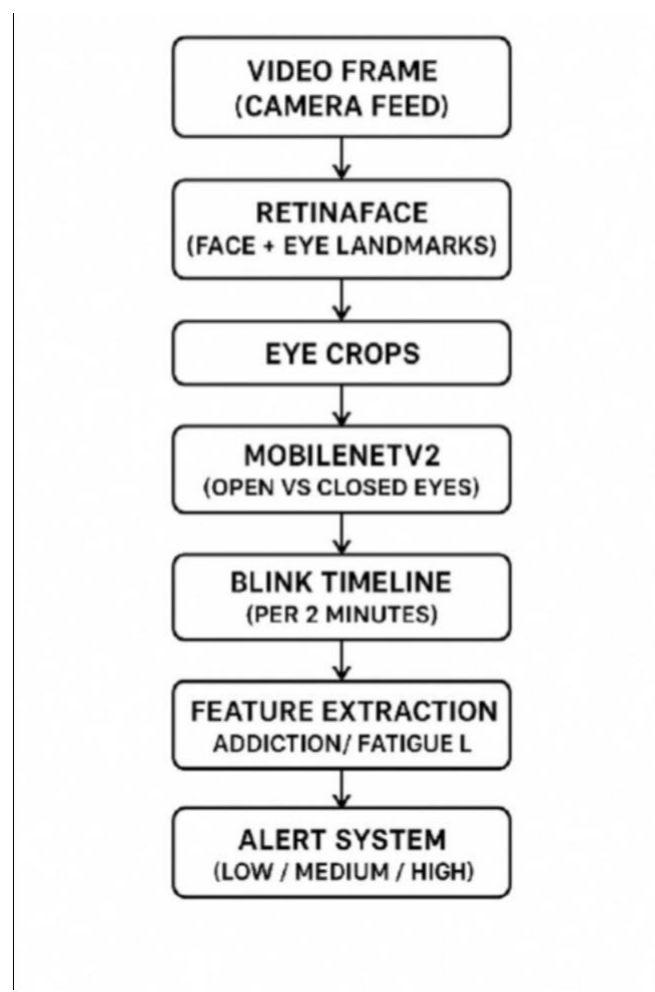


Figure 1: FLOW CHART

Step 1: VIDEO FRAME (CAMERA FEED)

This system starts with a live video that has been taken using a web camera or a smart phone camera. Such frames are the visual real time input of the eyes of the user which is further analyzed.

Step 2: RETINAFACE (FACE + EYE LANDmarks)

RetinaFace recognizes the face of a user and it manages to find the dots of the eye per frame. This enables the system to identify and isolate the parts of the eyes.

Step 3: EYE CROPS

The system simply crops the area of the eye which is identified. These are cut-out images that chop off the background noise and space required to perform analysis on blinking images.

Step 4: MOBILENETV2(OPEN vs Closed Eyes)

The mobile net V2 model is used to feed all the eye crops and determine whether the eye is open or closed. It triggers the series of eye states in real time that are frame by frame.

Step 5: BLINK TIMELINE (PER 2 MINUTES)

The eye-state results are also tabulated in two minutes timeline to monitor the trend of the blink like frequency of blink, duration of the blink, and irregularity. This is the chronology, the behavioural basis of addiction.

Step 6: 2D EXTRACTION OF FEATURES (ADDICTION / FATIGUE LEVEL)

The system computes such features as blink rate, variability, and eye-closures time according to the blink timeline. These indicators allow determining the degree of burnout or overuse of the smartphone by the user.

Step7: ALERT SYSTEM (LOW / MEDIUM / HIGH)

The system forecasts the addiction risk as low, medium or high depending on the extracted features. The user is informed to offer him/her guidelines or suggestions.

6.4 SYSTEM DESIGN

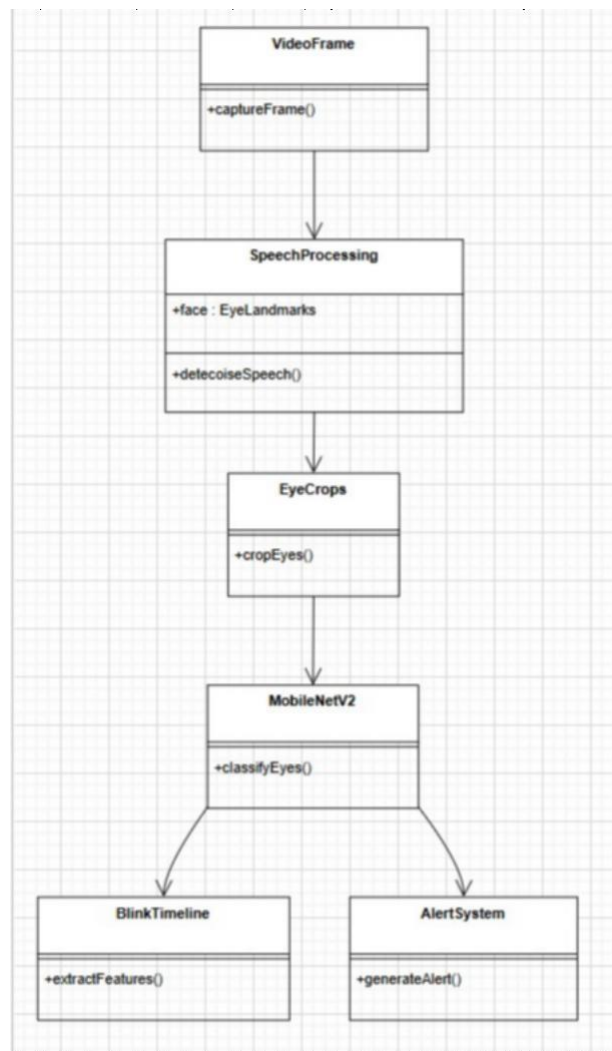


Figure 2: Class Diagram

System design defines a complete pipeline of the inputs (real time video) and the output (smartphone addiction level). The videoFrame module comprises the beginning of the process which involves the continuous frame of the webcam of the user. The frames are forwarded to the Face Processing module where the RetinaFace recognizes the face and locates correct eye markings.

The areas of the eye that are located are then forwarded to the EyeCrops component that isolates and cuts the area of the eye which will be subjected to further analysis. The MobileNetV2 model establishes the crops as an open or a closed eye state. These types form a series of time-serial that is analyzed by the BlinkTimeline module to obtain major blink features such as blink rate, blink duration and irregularity.

Finally, the AlertSystem module would take into consideration the optimized features

and provide a personalized addiction-risk alert (Low, Medium, or High), and provide the user with the immediate feedback. It is a structured design that is modular and enables appropriate detection, easy maintenance and real time monitoring without any difficulty.

6.5 ALGORITHMS IMPLEMENTED IN THE SYSTEM

- RetinaFace
- MobileNetV2 CNN
- Multi-Layer Perceptron (MLP)

RetinaFace: Retina face is a face detection model which is very complex and operates in single stage where the face model is identified and the facial features are recognised within a real time scenario. RetinaFace is based on the RetinaNet architecture and employs contextual facial features and dense regression to offer high accuracy levels even in difficult circumstances such as low-light, partial occlusion, small face size, and different head poses.

Contrary to typical multi-stage-based detection systems (detect a face and then apply separate models to predict landmarks), RetinaFace conducts face and landmark localization in a single forward pass, and hence, requires much fewer computational resources. It determines major landmarks such as eyes, nose tip, corners of the mouth, etc. very precisely.

RetinaFace is a very important tool in this project to be able to accurately determine the face in every frame taken by the camera and determine the specific areas of the eye with landmark coordinates. These clarity cut eye patches are the basis of proper classification of eye conditions and RetinaFace is an important part of a powerful and effective blink-based tracking and addiction discovery.

MobileNetV2: MobileNetV2 is a gentle convolutional neural network model, which is optimised to be executed on mobile and other resource constrained devices and environments. It introduces two significant innovations inverted residual blocks, and linear bottlenecks that cause 26 significant computational savings and maintain high precision. These design concerns allow MobileNetV2 to operate efficiently on smartphones and tablets and embedded systems without having to depend on a potent and powerful GPUs.

In the project, MobileNet V2 is used to classify all pictures of eye-region as an open or a closed picture. It is quick and efficient and can be applied in applications that require real-time with little delay between frame capture and prediction. The model uses every part of a cropped eye and gives the classification with a very low latency that helps measure eye

behavior such as blink duration and blink frequency and has the ability of constantly observing eye behavior.

As smartphone addiction and fatigue monitoring have their foundation in real-time monitoring of eye movement pattern, the rate of high inference of MobileNetV2 is referred to the possibility to execute the task with ease and high accuracy, including, detecting a blink on the low-power devices.

Multi-Layer Perceptron (MLP): Multi-Layer Perceptron (MLP) is a type of computer based artificial neural network; it comprises of numerous layers, one of which is an input layer, one or more hidden layers and an output layer. Each layer is characterized by nonlinear transformation with the help of the activation functions and it allows the network to learn complex relations between features and output classes.

The MLP classification used in the current project operates on the statistics features of blink which are extracted over a given time window i.e. such as:

- Average blink rate
- Maximum eye-open duration
- Variance in blink intervals
- Ocular stress in duration and intermittency of blink.

The traits are projections of physiological and cognitive trends of smartphone addiction or exhaustion. The MLP is conscious of the pattern of discrimination of these traits to classify the extent of addiction or fatigue of the user as low, medium, or high risk. 27 MLPs are also fast and efficient in their computation and hence can easily be used on mobile devices whose resources are few. The final decision making layer in the system is the MLP model which converts real time information of the blink into meaningful behavioural understanding with a high degree of accuracy.

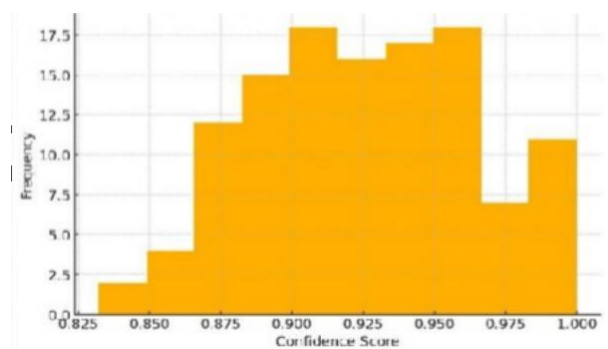


Figure 3: RetinaFace Detection Confidence Distribution

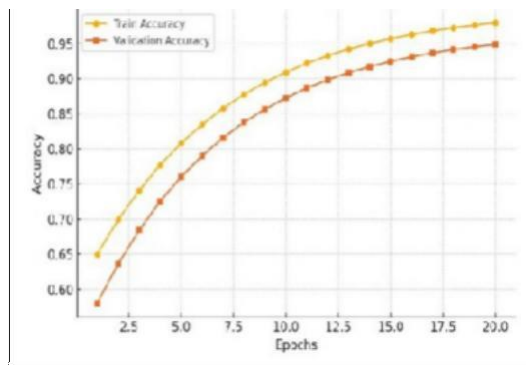


Figure 4: MobileNetV2-Accuracy over Epochs

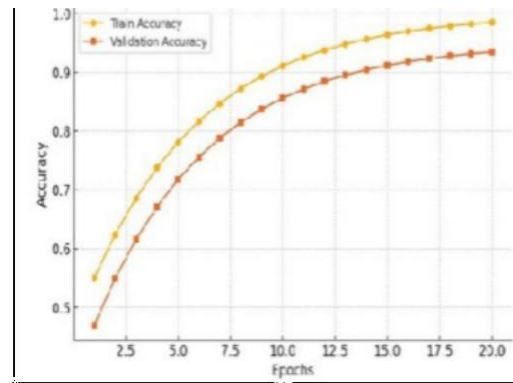


Figure 5: MLP Classifier

Final Accuracy Table

Model	Training Accuracy	Validation Accuracy
RetinaFace	N/A (not a classifier)	0.93 (avg-detection confidence)
MobileNetV2 (CNN)	0.98	0.95
MLP (Blink Features)	0.99	0.93

CHAPTER 7: RESULT AND DISCUSSION

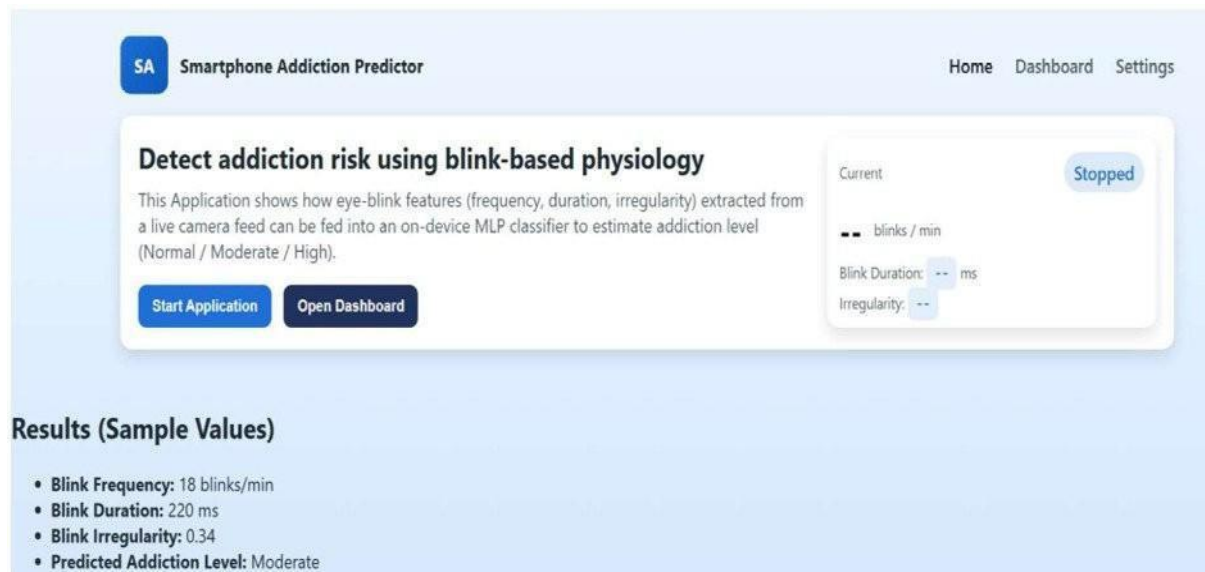


Figure 6: Home Page

The Smartphone Addiction Predictor, the Home Page offers the general information about the system and enables users to begin with the real-time analysis based on the blink. It shows the choice of options to start the application or open the dashboard and a clean interface with the values of the current blink status, the blink duration and irregularity. Sample outcomes can include the frequency of blink, the length of the blink, the irregularity of the blink and the expected level of addiction which will help a user know how the system perceives the eye-blink patterns. The interface will be easy to navigate with simplicity and clarity to its users who want to monitor their risk of addiction to their smartphones.

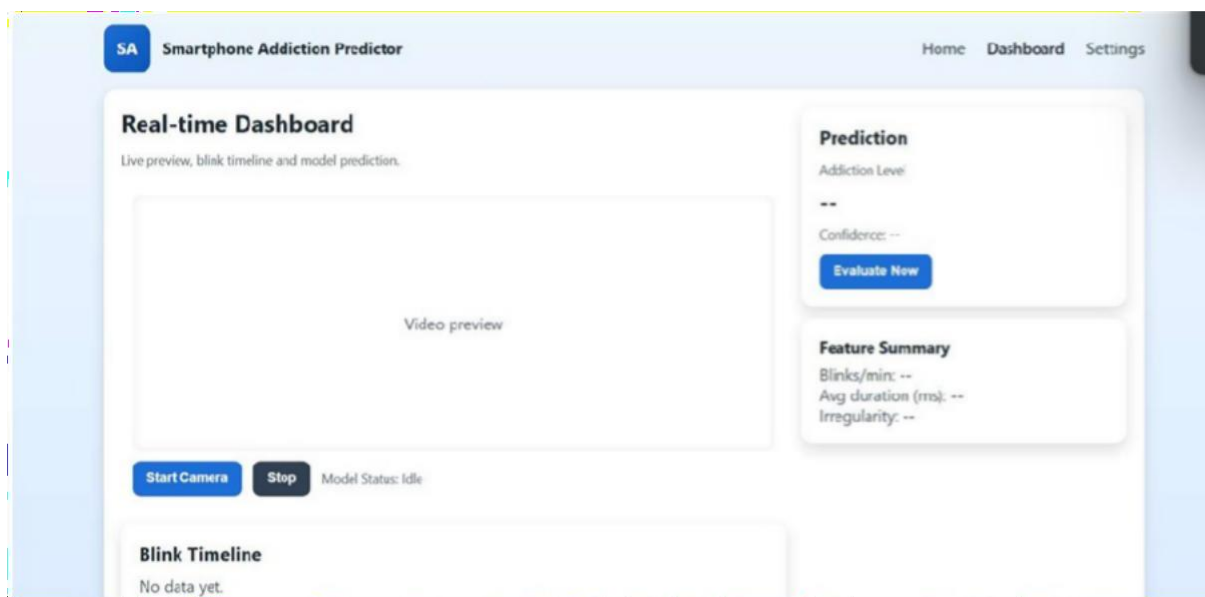


Figure 7: Dashboard Page Before Starting Application

Before the application has been started, an empty video preview area and prediction results and feature summary placeholders are displayed on the dashboard. The user can also start the camera to start real-time blink recognition and view the subsequent timeline and model results. Here, the blink information and addiction prediction are not shown and the state of the model is idle. In order to obtain the determinacy of the interface, the demonstration of what will be monitored after the system is turned on will be provided and this will help users to acquire knowledge about the workflow before the real-time analysis can begin.

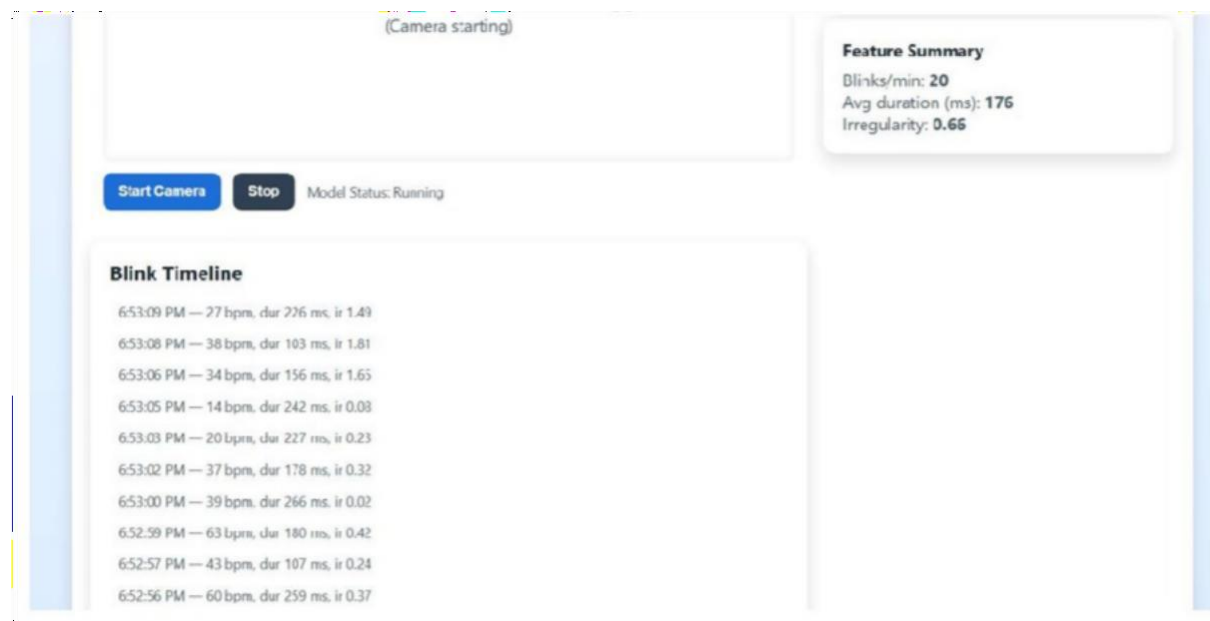


Figure 8 : Blink Time

The Blink Timeline section shows a live log of blink data recorded as the camera is operating and this includes the frequency of blink (bpm), the length of the blink, and the irregularity of the blink. The stream is a continuous one that assists the user to see how his or her blinking behaviour evolves with time when using the smartphone. The feature summary is updated automatically on the right side, and it gives an immediate summary of the physiological state of the user in overall averages. The view aids the system to examine behavioural patterns that contribute to prediction on addiction level.

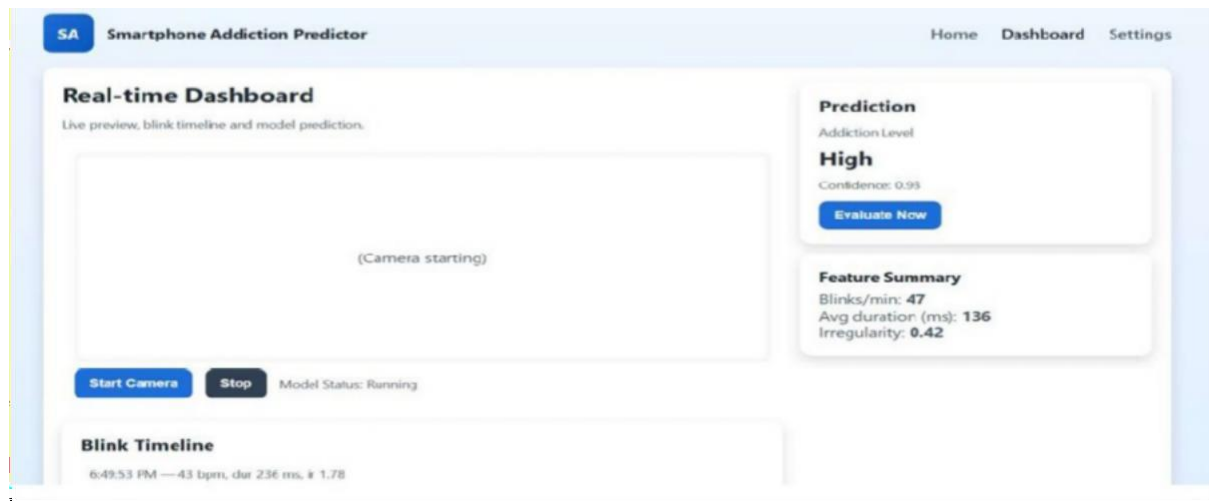


Figure 9: Predicting High

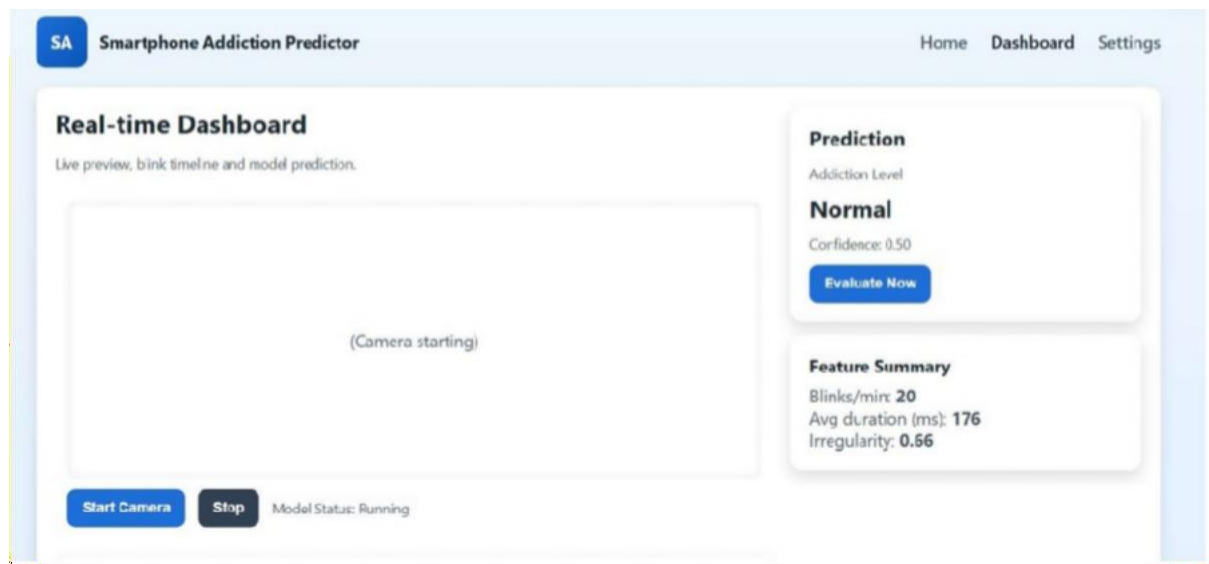


Figure 10: Predicting Normal

CHAPTER 8 : CONCLUSION

The developed system successfully predicts smartphone-induced fatigue by analyzing blink behavior using a deep learning–based pipeline, proving that eye activity is a strong and reliable indicator of digital overuse and cognitive tiredness. The integration of RetinaFace for highly accurate facial and eye landmark detection, MobileNetV2 CNN for classifying eye state (open or closed), and a Multi-Layer Perceptron (MLP) for fatigue level prediction enables the model to operate efficiently and non-intrusively in real time. Through continuous frame capture and preprocessing (grayscale conversion, histogram equalization, Gaussian smoothing, normalization, and resizing), the system extracts a clean and consistent Region of Interest (ROI) from the eye. MobileNetV2 then analyzes these eye crops frame-by-frame and classifies each as open or closed, after which blink patterns are observed over a sliding time window.

Statistical blink-based features such as blink rate, blink interval variance, longest eye-open duration, and cumulative session duration are calculated and input to the MLP to determine fatigue levels (Low, Medium, or High). With 95% classification accuracy for MobileNetV2 and 93% fatigue 33 prediction accuracy using MLP, the results confirm that automated blink monitoring is a dependable measurement of fatigue and attention decline during prolonged smartphone usage.

The system’s high accuracy and real-time performance demonstrate its potential applications in digital wellness, safe driving alerts, online learning monitoring, workplace productivity, and preventive warning systems for eye strain and smartphone addiction. Overall, this project proves that deep learning combined with physiological visual cues can accurately detect fatigue without physical sensors or user interaction, offering a practical solution to promote healthier digital habits and reduce the negative effects of excessive screen exposure.

CHAPTER 9: FUTURE SCOPE

The proposed algorithm of the digital wellness monitoring as the smartphone addiction prediction system is a solid starting point to do the real-time and can be significantly expanded in the future research. One of the main directions is the establishment of a complete Android and iOS application in a mobile version, which will provide the ability to track addictions constantly and on the phone without resorting to external mechanisms. With the introduction of this technology in phones it would see millions of users in the globe access digital wellness monitoring.

The other important enhancement to make in the future is the application of more advanced deep learning models such as LSTM, GRU or Transformer models. This type of model can make long-term dependencies on the blink patterns, with the help of which it is possible to make more exact and contextual forecasts regarding the level of addiction. This would also facilitate the system to be more accommodative to the different age groups, lighting and behaviors of the users.

Another area that is ready to grow is multimodal data fusion since the blink activity can be complemented with other signals of the user. It could be improved in the future to check head position, or pupil dilation, or facial expressions of stress, or even have data about smartwatches 34 like heart rate variability or sleep patterns. A combination of a few physiological and behavioural signs would create a more complete and accurate diagnosis of digital fatigue and addiction.

The system can also evolve to a personalized digital wellness coach whereby the system will make personalized recommendations beginning with the behavior of the user. It may suggest the times of the breaks, the limitations of the screen time, or relaxation exercises and generate weekly/monthly reports about the eye health and attention improvement. This would contribute to transforming the system into a more of an active digital health assistant, opposed to a monitoring tool.

The second direction that is also significant in the future is the long-term behavioral analytics where the system will keep a history of trends and patterns over time. It would allow the user to understand the dynamics of growth in the degree of their addiction and, also, allow them to determine what factors make them more dependent on their smartphone.

Along with personal applications, the mental health monitoring has a high potential of the system as well as the stress, the cognitive load, and fatigue in the eye can be all associated with the rate of blinking and eye activity. The tool would be beneficial to mental health

professionals as it would help in the early identification of the symptoms of anxiety, burnout, or depression.

The system could be implemented on a corporate level, such as in rising the productivity of the staff people who work long shifts in front of the screens; the system would be employed to identify the digital fatigue of the staff people. It can support organizations to be healthier by working and going digital.

Parental control applications can be listed among the most significant benefits to the society, as the system can help parents reveal the harmful tendencies of children and teenagers using the devices. This can be achieved by giving them timely warnings to guide them toward moderated screen time and prevent future troubles.

With little modification to the same eye-tracking pipeline, fatigue among drivers in cars can be detected so that road safety may be ensured by identifying drowsiness at a very early stage. The simple technology can be enabled to permit several real time monitoring applications in industries.

Finally, both the IoT devices and wearable technology can be linked to the future versions to receive more detailed and continuous physiological data. The integration of smart glasses, VR gadgets or internet of things health sensors can create a net of digital wellness devices all operating on the same simple deep-learning platform.

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